MIL-S-24298A(SHIPS)
12 February 1969
SUPERSEDING
MIL-S-24298(SHIPS)
9 May 1967
(See 6.4)

MILITARY SPECIFICATION

SAFETY BELT AND HARNESS, SUBMARINE PERSONNEL

1. SCOPE

1.1 Scope. This specification covers safety belt and harness and attachments (referred to as "safety belt") for the protection of personnel aboard submarines.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
- V-T-295 - Thread, Nylon.
- QQ-B-726 - Bronze Castings, Manganese and Aluminum Manganese.
- QQ-P-416 - Plating, Cadmium (Electrodeposited).
- QQ-S-698 - Steel, Sheet and Strip, Low-Carbon.
- DDD-L-20 - Label; for Clothing; Equipage and Tentage (General Use).

MILITARY
- MIL-F-495 - Finish, Chemical, Black, for Copper Alloys.
- MIL-W-8630 - Webbing, Textile, Nylon, Latex Impregnated.
- MIL-P-16232 - Coatings, Phosphate, Heavy (Manganese and Zinc Type) and Phosphate Treating Solutions.
- MIL-C-21768 - Copper Alloy Numbers 210 (Gilding, 95 percent) and 220 (Commercial Bronze, 90 percent) Sheet and Strip.

STANDARDS

FEDERAL
- FED-STD-151 - Metals; Test Methods.
- FED-STD-751 - Stitches, Seams and Stitchings.

DRAWING
NAVAL SHIP ENGINEERING CENTER
- 805-2482847 - Safety Belt and Harness, Submarine - Detailed Assembly.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
- Uniform Freight Classification Rules.

(Applications for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production a sample shall be tested as specified in 4.2 (see 6.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:55Z
Check the source to verify that this is the current version before use.

3.2 Design, material, and fabrication. The design, material, and fabrication of the safety belt and harness shall be in accordance with Drawing 805-2482847 and this specification. In the event of conflict, this specification shall govern.

3.2.1 Webbing. The nylon webbing used in the safety belt shall be 2 inches wide and shall conform to type II of MIL-W-8630. Unless otherwise specified (see 6.1), color shall be white. The ends of all webbing shall be smoothly fused to prevent raveling.

3.2.2 Thread. All thread used in the construction of the harness and belt shall be nylon and shall conform to V-T-295. Thread for the shoulder harness shall be size F-P, class 1 with a minimum breaking strength of 16 pounds. Thread for the waist band shall be size S, class 1 with a minimum breaking strength of 42 pounds. The color shall suitably match the color of the belt and harness.

3.3 Hardware.

3.3.1 Metal parts. All metal parts of the safety belt and harness shall withstand the tests of 4.5.1.1 and 4.5.1.2.

3.3.2 Waist belt clasp. The clasp used to secure the waist belt shall conform to Hollander Metal Products Company part no. SP-514, or equal, and shall be fabricated of manganese bronze conforming to QQ-B-726. The clasp shall have a lusterless black chemical finish conforming to MIL-F-495.

3.3.3 Waist belt slide. The slide adjuster, used to adjust the length of the waist belt, shall be made of drop-forged steel and shall conform to North and Judd part no. M-3359, or equal.

3.3.4 D-ring. The safety D-ring, shall conform to R.H. Buhrke Company, part no. 2038, or equal and shall be fabricated from forged carbon steel (1025).

The D-ring shall be cadmium plated in accordance with class 1 of QQ-P-416 with a supplementary phosphate treatment in accordance with MIL-P-16232. The phosphate coating shall be 0.0008 to 0.0010 inch thick.

3.3.4.1 Wear tube. A sheet steel wear tube (18 to 20 gage steel) in accordance with QQ-S-698 cold rolled, commercial quality, shall form a sleeve over the crossbar of the D-ring. The diameter of the tube shall be approximately 1/2 inch to permit the tube to turn freely around the crossbar, and the length shall be 1/16 inch less than the inside width of the crossbar (1-7/8 inches). The tube shall be cadmium plated in accordance with class 1 of QQ-P-416 with a supplementary phosphate treatment in accordance with MIL-P-16232. The phosphate coating shall be 0.0008 to 0.0010 inch thick.

3.3.5 Shoulder strap slide. The slides used for the loop adjustments on the shoulder strap shall conform to Hollander Metal Products Company, part no. SP-348, or equal. The slides shall be bronze stamping in accordance with MIL-C-21768 and shall have a lusterless black finish conforming to MIL-F-495.

3.3.6 Label. Each belt assembly shall have an identification label measuring approximately 1-1/2 inches wide by 2-3/4 inches long. The labels shall conform to type I, class I of DDD-L-20. Unless otherwise specified (see 6.1), labels shall have the following inscription and size of characters:

Inscription	Size of Characters
Safety Belt and Harness	18-point Gothic Bold
Submarine Personnel	18-point Gothic Bold
Contract or order number	10-point Gothic Medium
Date of Manufacture	10-point Gothic Medium

3.3.7 Stitches. All stitches shall be in accordance with stitch type 301 of FED-STD-751. Unless otherwise specified, all stitching shall be accomplished using a cross-box stitch with a double stitch at either end across the web and the outer edge stitching shall be 1/8 inch gauge from the outer edge. The double stitch at either end of the box across the web shall be approximately 1/8 inch apart.

3.3.7.1 Stitches per inch. The number of stitches shall be 8 to 12 stitches per inch, and shall be interpreted to mean the minimum and maximum number permitted.

3.3.8 Waist belt. The waist belt shall be made up of webbing as specified in 3.2.1. The cut length of 2 inch wide nylon webbing shall be 65-1/2 plus or minus 1/4 inches long. The male portion of the clasp (see 3.3.2) and the D-ring (see 3.3.4) shall be secured to one end of the belt by looping the end of the webbing through the clasp, back over and through the D-ring with the webbing end between the webbing going and coming from the clasp and stitching. On the opposite end, the female portion of the clasp (see 3.3.2) and the slide (see 3.3.3) shall be attached.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:55Z
Check the source to verify that this is the current version before use.

3.3.9 Shoulder harness. The shoulder harness shall consist of two pieces of 2-inch wide nylon webbing, each having a cut length of 43 plus or minus 1/4 inches. One end of each piece of webbing shall be threaded through the two slides specified in 3.3.5, turned under approximately 2-1/4 inches and sewn with a modified cross-box stitch, 1-1/4 inches long by 1-3/4 inches wide. The opposite end of each piece shall be turned under approximately 4-1/2 inches and similarly sewn to form a loop approximately 3 inches long. The individual harness straps shall be joined to each other at an angle of 60 degrees and approximately 7-1/2 inches from the permanent 3-inch loop.

3.4 Identification markings. Each safety belt shall have, securely sewn thereto, a label as specified in 3.3.6. Location of the label shall be as detailed in Drawing 805-2482847.

3.5 Workmanship. The workmanship shall be in accordance with highgrade manufacturing practice covering this class of work.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 When inspection is conducted at the contractor's plant, all tests specified herein shall be conducted by the contractor under the surveillance of the inspector. Contractors not having laboratory facilities satisfactory to the inspector shall engage the services of a commercial testing laboratory acceptable to the inspector. The contractor shall furnish tests reports, in duplicate, showing quantitative results of all tests and analysis required by this specification, and signed by the Director, or his authorized assistant, of the laboratory in which the tests were conducted. These reports shall be furnished to the inspector. The furnishing or maintaining of test reports or records shall not be construed as waiving the witnessing of tests by the inspector. Source inspection of sub-contracted material to insure its meeting the requirements of this specification may be utilized by the inspector cognizant of the prime contract or order, at his discretion. Previous acceptance or approval of material shall not be construed as a guarantee of the acceptance of the finished product.

4.2 First article inspection. First article inspection shall consist of all the examination and tests specified in 4.4 and 4.5.

4.3 Sampling for quality conformance inspection.

4.3.1 Lot. All belts or attachments presented at one time shall be considered a lot for purposes of inspection.

4.3.2 Sampling for visual and dimensional examination. A random sample of safety belts or attachments shall be selected in accordance with table I for the examination specified in 4.4.1.

Table I - Sampling for visual and dimensional examination.

Number of belts or attachments in inspection lot	Number of belts or attachments in sample	Acceptance number (defectives)	Rejection number (defectives)
25 or under	10	0	1
26 to 65	15	0	1
66 to 110	25	1	2
111 to 300	35	1	2
301 and over	50	2	3

4.3.3 Sampling for tests. A random sample of safety belts or attachments shall be selected in accordance with table II for the tests specified in 4.5.

Source: https://assist.dla.mil -- Downloaded: 202640-19T20:55Z
Check the source to verify that this is the current version before use.

Table II - Sampling for quality conformance tests.

Number of belts or attachments in lot	Number of belts or attachments in sample
40 and under	2
41 to 110	3
111 to 300	4
301 and over	5

4.4 Quality conformance inspection.

4.4.1 Visual and dimensional examination. Each of the sample safety belts and attachments selected in accordance with table I shall be visually and dimensionally examined to verify conformance to this specification. Examination shall be for defects as specified in table III.

Table III - Classification of defects.

Categories	Defects
Critical:	
1	Buckle parts not mating, locking, or unlocking with facility.
2	Nylon webbing, ragged or loose strands, ends not properly heat sealed.
3	Slides not holding webbing or causing webbing to bind.
4	Intertwined, tangled, or bunched threads in seams.
5	Sample safety belt does not pass tensile test of 4.5.1.2.
Major:	
101	Incomplete; any component parts missing.
102	Material not as specified.
103	Parts or size not specified on drawing.
104	Metal parts not free of burrs and sharp edges.
105	Parts not in accordance with approved drawing.
106	Slides not operating smoothly.
107	Stitching not conforming to the requirements of this specification.
108	Stitches not firmly in place, not evenly spaced and dimensions incorrect.
109	Not proper number of stitches per inch.
Minor:	
201	Color not as specified or uneven shade.
202	Thread ends not snipped.

4.4.1.1 Rejection. Any belt or attachment containing one or more defects shall not be offered for delivery and if the number of defective belts and attachments in the sample exceeds the acceptance number for that sample, this shall be cause for rejection of the lot represented by the sample.

4.4.2 Quality conformance tests. Each of the samples selected in accordance with 4.3.3 shall be tested as specified in 4.5.1 and 4.5.2 (see 4.5.3). Failure of any sample to pass the tests of 4.5.1 and 4.5.2 shall be cause for rejection of the lot represented by the sample.

4.5 Test procedures.

4.5.1 Drop test and tensile test. After samples of safety belts or harnesses have been subjected to the tests of 4.5.1.1 and 4.5.1.2, the samples shall be destroyed.

4.5.1.1 Drop test. Securely fasten the belt around a 300 pound bag of sand, then connect by means of a snap hook, a length of suitable rope line having a minimum breaking strength of 8,400 pounds to the D-ring on the belt. Secure the free end of the rope line to a rigid support. Allow the bag to fall freely through a distance of 6 feet. Repeat the test two more times without removing the belt from the bag. Examine the safety belt for any rupture of webbing or thread or deformation of metal components to determine conformance with requirements of this specification.

Source: https://assist.dla.mil -- Downloaded: 20260-19T20:55Z
Check the source to verify that this is the current version before use.

4.5.1.2 Tensile test. The waist belts which have successfully completed the test specified in 4.5.1.1 shall be subjected to the following test. With the clasp fastened in the service position, the belt shall be placed over 2-inch mandrels. The mandrels shall be separated at a uniform rate of 2 plus or minus 1 inches per minute. The pull on the belt shall be such that the clasp does not contact with either mandrel and only one thickness of webbing passes over each mandrel. Deformation of metal parts or parting of webbing material shall not occur under 2,000 pounds and shall not fail under a total load of 4,000 pounds.

4.5.2 Salt spray. A 100-hour salt spray test shall be conducted on all the metal components in accordance with method 811 of FED-STD-151. There shall be no evidence of corrosion which will adversely affect the operation of the equipment.

4.5.3 Tensile completion test. After the lot has passed the drop test and tensile test of 4.5.1.1 and 4.5.1.2, each safety belt in that lot shall be subjected to a 1,500 pound tensile completion test.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.2.)

5.1 Packaging. Packaging shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early use and may conform to the supplier's commercial practice when such meets these requirements.

5.1.1 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity for early use. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation and may conform to the suppliers commercial practice when such meets these requirements. Each safety belt shall be packed as a unit.

5.1.2 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractors commercial practice. The information shall include nomenclature, Federal stock number or manufacturer's part number, contract or order number, contractor's name and designation.

5.2 Domestic shipment and storage or overseas shipment (see 5.3). The requirements and levels of packaging, packing and marking for shipment shall be specified by the procuring agency (see 6.1).

(5.2.1 The following provides various levels of protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Packaging, packing and marking (see 5.2). Safety belts shall be packaged level A or C, packed level A, B or C as specified (see 6.1), and marked in accordance with MIL-P-3454.)

5.3 Use of polystyrene (loose-fill) material.

5.3.1 For domestic shipment and early equipment installation and level C packaging and packing. Unless otherwise approved by the procuring activity (see 6.1), use of polystyrene (loose-fill) material for domestic shipment and early equipment installation and level C packaging and packing applications such as cushioning, filler and dunnage is prohibited. When approved, unit packages and containers (interior and exterior) shall be marked and labelled as follows:

"CAUTION

Contents cushioned etc. with polystyrene (loose-fill) material.
Not to be taken aboard ship.
Remove and discard loose-fill material before shipboard storage.
If required, recushion with cellulosic material bond fiber,
fiberboard or transparent flexible cellular material."

5.3.2 For level A packaging and level A and B packing. Use of polystyrene (loose-fill) material is prohibited for level A packaging and level A and B packing applications such as cushioning, filler and dunnage.

Source: https://assist.dla.mil -- Downloaded: 20261O-19T20:55Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Whether first article sample is required (see 3.1).
(c) Color required for safety belt (see 3.2.1).
(d) Inscription and size of character of labels, if different (see 3.3.6).
(e) Packaging, packing or marking requirements other than those required by 5.1, 5.2 and 5.3.
(f) Approval for the use of polystyrene material (see 5.3).

6,2 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.3 First article. Invitation for bids should provide that the Government reserves the right to waive the requirements for samples of first article as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.4 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 4240-N351)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:55Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
NAVAL SHIP ENGINEERING CENTER
WASHINGTON, D. C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

COMMANDER, NAVAL SHIP ENGINEERING CENTER
DOD STANDARDIZATION PROGRAM AND DOCUMENTS BRANCH
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:55Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

(NAVSHIPS 0VPRT 12-66)

Source: https://assist.dla.mil -- Downloaded: 202610-19T20:55Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T20:55Z
Check the source to verify that this is the current version before use.